

Motorcyclist Helmet-Use Detection in Traffic Images Using Fine-Tuned YOLOv8m and Rule-Based Spatial Reasoning

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Abstract Motorcycle helmet use is a critical road-safety factor, yet image-based helmet monitoring requires more than isolated helmet detection: helmets must be associated with the correct motorcycle riders. This paper presents an interpretable computer-vision pipeline for rider-level helmet-use detection in traffic images. A pretrained YOLOv8m detector is fine-tuned to detect three primitive object classes—person, motorcycle, and helmet—and a deterministic spatial-reasoning module then associates persons with motorcycles and helmets to infer rider-helmet and rider-no-helmet labels. The selected 120-epoch YOLOv8m run achieved a validation mAP@0.5 of 0.9190, precision of 0.8740, recall of 0.8719, and mAP@0.5:0.95 of 0.6905 on the three-class detection task. The study emphasizes a transparent detection-and-association formulation and clearly separates validated detector performance from the still-unmeasured final rider-level classification stage.

Keywords: helmet detection, traffic safety, YOLOv8m, object detection, spatial reasoning, intelligent transportation systems

1 Introduction

Motorcycles are widely used in many developing countries, including Vietnam. Road-safety reports indicate that traffic accidents remain a major global public-health problem [1]. For motorcycle riders, helmet use is one of the most important protective measures, and prior evidence shows that helmets reduce the risk of head injury and death [2]. Automated helmet-use monitoring can therefore support traffic-safety supervision, compliance analysis, and intelligent transportation applications.

Modern object detectors, especially the YOLO family, provide an attractive speed–accuracy trade-off for efficient visual recognition [3, 4]. However, helmet-use detection is not equivalent to detecting helmets in isolation. A detected helmet must be associated with the correct motorcycle rider. Pedestrians wearing helmets, helmets carried by other people, or helmets attached to parked motorcycles can otherwise lead to incorrect conclusions.

This study investigates an interpretable detection-and-association pipeline for rider-level helmet-use recognition. A pretrained YOLOv8m detector is fine-tuned to detect three primitive classes: person, motorcycle, and helmet [5, 6]. The raw detections are then fused by deterministic geometric rules to infer rider-level helmet status. The

central focus is not only detecting helmets, but also making the association step explicit, auditable, and separable from the learned detector.

The main contributions are as follows:

- An interpretable rider-level helmet-use detection pipeline that combines YOLOv8m primitive detections with explicit spatial association rules.
- A dataset preparation workflow that merges Pascal VOC and YOLO annotations into a consistent three-class detection format.
- A validation analysis of the fine-tuned detector, including precision, recall, mAP@0.5, and mAP@0.5:0.95 across complete training runs.
- A limitation-aware evaluation protocol that distinguishes measured detector performance from the unevaluated final rider-level labels.

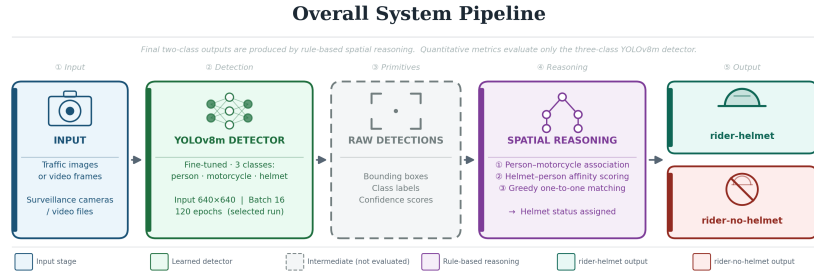


Fig. 1 Overall pipeline for rider-level helmet-use detection. The detector predicts primitive person, motorcycle, and helmet boxes; deterministic spatial reasoning then associates these detections to infer rider-helmet or rider-no-helmet labels.

2 Related Work

2.1 Single-Stage Object Detection

YOLO formulates object detection as a single-stage prediction problem and has been widely used in efficient visual-recognition applications [3]. YOLOv4 further improved the accuracy–speed trade-off through architectural and training improvements [4]. YOLOv8, released by Ultralytics, provides modern detector variants at different model scales; YOLOv8m offers a practical compromise between accuracy and computational cost for the present study [5, 6].

Table 1 Main dataset split used by `data.yaml`.

Split	Images	Approx. ratio
Train	917	80%
Validation	114	10%
Test	116	10%
Total	1,147	100%

2.2 Helmet-Use and Traffic-Safety Detection

Helmet-use detection has been studied using surveillance videos and deep-learning detectors. Dahiya et al. proposed automated detection of bike riders without helmets in surveillance videos, emphasizing the need to interpret helmet status in the context of motorcycle riders [7]. Recent studies have explored YOLO-based helmet violation detection and few-shot data sampling for helmet-related traffic monitoring [8, 9]. The present study follows this application direction while separating primitive object detection from rider-level spatial association.

2.3 Spatial Reasoning After Detection

Object detectors typically output independent bounding boxes and confidence scores. For helmet-use monitoring, these primitive detections must be combined into rider-level decisions. The proposed pipeline therefore treats person, motorcycle, and helmet detections as intermediate evidence and uses explicit spatial constraints to associate them. This design improves interpretability and makes the remaining evaluation gaps easier to identify.

3 Dataset and Preprocessing

The main dataset contains 1,147 images from two sources. The first source is a public helmet dataset with 764 images annotated in Pascal VOC format. The second source is a custom traffic dataset with 383 images annotated in YOLO format. The annotations were converted and merged into a consistent YOLO-style structure before training.

The main experiments used the split specified in `data.yaml`. The split was generated randomly with seed 42 and an approximate 80/10/10 ratio. Table 1 summarizes the image counts. Because several images are extracted from videos or similar locations, visually similar frames may appear in different splits. This introduces a potential data-leakage risk and motivates future splits by video or location.

A separate `dataset_combined.yaml` configuration included 753 additional auto-labeled frames from video, producing 1,214 training images and 303 validation images

without a test split. However, the corresponding combined runs failed early and are not used as main results.

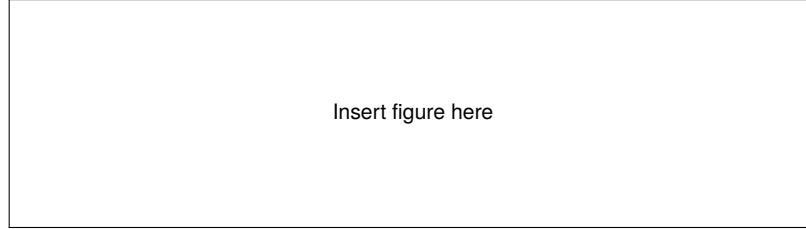


Fig. 2 Space reserved for representative dataset samples and annotations. This figure should show examples from the public Pascal VOC source and the custom YOLO-format traffic-image source.

Preprocessing included XML-to-YOLO annotation conversion, folder normalization, image resizing, and label inspection. Training-time augmentation included horizontal flipping, rotation or affine transformations, HSV and brightness changes, Gaussian blur, Gaussian noise, and mosaic augmentation.

4 Method

4.1 Problem Formulation

Given an input traffic image, the goal is to assign helmet-use status at the rider level rather than merely detect helmets as independent objects. The pipeline therefore decomposes the task into two stages. First, a detector predicts primitive bounding boxes for persons, motorcycles, and helmets. Second, a spatial-reasoning module groups these primitive detections into rider candidates and assigns a final semantic label. A rider candidate is labeled as rider-helmet when a helmet is associated with the corresponding person; otherwise, the candidate is labeled as rider-no-helmet.

4.2 Three-Class Detector

The learned component is a pretrained YOLOv8m detector fine-tuned for three primitive classes: person, motorcycle, and helmet. The model receives a resized traffic image and outputs bounding boxes, confidence scores, and class labels. This formulation keeps the neural detector focused on local visual evidence while leaving the rider-level interpretation to a transparent post-processing stage. Such separation is useful for error analysis because missed primitive detections and incorrect spatial associations can be inspected independently.

4.3 Rider Candidate Construction

Rider candidates are formed by associating detected persons with nearby motorcycles. For each person–motorcycle pair, the algorithm considers geometric compatibility based on bounding-box overlap, the position of the person relative to the motorcycle, and the distance between box centers. Candidate pairs that satisfy the spatial constraints are retained as potential riders. This step reduces false helmet-use decisions caused by pedestrians, parked motorcycles, or helmets that are visible in the scene but unrelated to an active rider.

4.4 Helmet Association

For each rider candidate, helmet detections are matched to the corresponding person box. The matching score combines five interpretable cues: whether the helmet center lies in the upper region of the person box, normalized overlap with the person region, distance to the estimated head area, height alignment, and helmet confidence. The affinity score is defined as

$$\text{Aff}(P, H) = 0.35C_{pos} + 0.15C_{overlap} + 0.30C_{dist} + 0.10C_{height} + 0.10C_{conf}. \quad (1)$$

A helmet is assigned to a rider when $\text{Aff}(P, H) \geq 0.5$. Greedy one-to-one matching ensures that each helmet is linked to at most one rider. The final rider label is then produced deterministically from the matched helmet evidence.

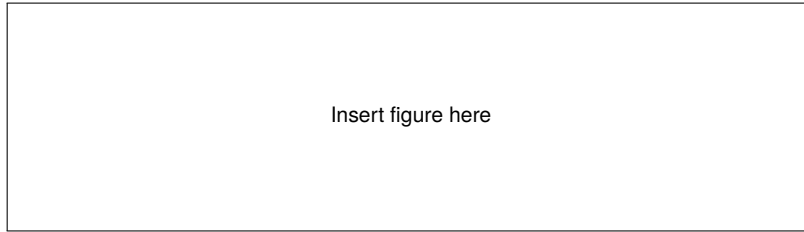


Fig. 3 Space reserved for the spatial-reasoning diagram. The final version should illustrate person–motorcycle association, helmet–person matching, and the rider-helmet/rider-no-helmet decision rule.

4.5 Training and Evaluation Protocol

The selected training configuration used YOLOv8m pretrained weights, an input size of 640, a batch size of 16, and 120 epochs. The reported quantitative results are computed on the validation split for the three primitive detection classes. The reserved test split

Table 2 Training configuration for the selected run.

Item	Value
Model	Pretrained YOLOv8m
Classes	person, motorcycle, helmet
Epochs	120
Input size	640
Batch size	16
Evaluation during training	Validation split
Final rider-level GT	Not available

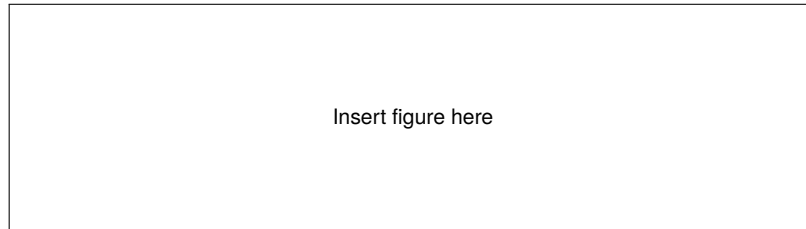
Table 3 Complete YOLOv8m validation runs for the three-class detection task.

Run	mAP50	Prec.	Rec.	mAP50-95
Run 1, 100 ep.	0.9219	0.8861	0.8473	0.6974
Run 2, 100 ep.	0.9194	0.8821	0.8360	0.6724
Run 3, 120 ep.	0.9190	0.8740	0.8719	0.6905
Run 4, 120 ep.	0.9074	0.8736	0.8450	0.6967

and the final rider-level labels are not treated as validated quantitative results because a separate test evaluation and rider-level ground truth are not yet available.

5 Results

Table 3 summarizes the four complete YOLOv8m validation runs. The selected 120-epoch run achieved mAP@0.5 of 0.9190, precision of 0.8740, recall of 0.8719, and mAP@0.5:0.95 of 0.6905. The highest mAP@0.5 among the four runs was 0.9219 at 100 epochs, while the selected 120-epoch run had the highest recall.

**Fig. 4** Space reserved for additional evaluation material, such as training curves, a validation confusion matrix, or representative qualitative predictions.

Qualitative testing revealed two main error sources. First, missed helmet detections can cause helmeted riders to be labeled as non-helmeted, especially when helmets are small, occluded, or captured under poor lighting, rain, fog, or long distance. Second, spatial post-processing can fail in crowded scenes with overlapping riders and motorcycles or helmets outside the expected head region.

6 Discussion and Limitations

The current study has several limitations. First, the main quantitative results are validation-only metrics; the independent test split has not yet been evaluated. Second, random splitting may introduce leakage because frames from the same video or location may appear in multiple subsets. Third, the final rider-helmet/rider-no-helmet labels do not have ground-truth annotations and were not evaluated using precision, recall, F1-score, or a confusion matrix. Fourth, no baseline or ablation study was conducted, so the contribution of the spatial-reasoning module cannot yet be quantified. Finally, all complete runs used YOLOv8m, without comparison to lighter or heavier variants.

7 Conclusion and Future Work

This paper presented an interpretable motorcyclist helmet-use detection pipeline based on fine-tuned YOLOv8m primitive detections and rule-based spatial reasoning. The detector achieved strong validation performance on the three-class detection task, while the final rider-level labels remain a separate target for future quantitative evaluation. Future work should evaluate the reserved test set, annotate rider-level ground truth, perform ablation studies for the spatial-reasoning module, and compare YOLOv8m with lighter and heavier detector variants.

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Competing Interests The author declares no competing interests relevant to the content of this paper.

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